

Numerical Analysis Exercise 11

Problem 1: Given the I.V.P.

$$y' = \frac{2}{x}y + x^2 e^x, 1 \leq x \leq 2, y(1) = 0$$

with the exact solution $y = x^2(e^x - e)$.

Use Euler method and Modified Euler method to get the numerical solution respectively, and compare your numerical solutions with the exact solution and demonstrate your comparisons by graph.

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%-----I.V.P-----  
%-----Initial-value problem-----  
%Euler method  
% We set the f(t_k,y(t_k)) as the y'; and we use the first-order Taylor  
% expansion, regard the h as the (x-x0).  
  
clc  
clear all  
close all  
  
a = 1;  
b = 2;  
M = 20;  
h = (b-a)/M;  
  
y_k = zeros(1,M);  
y_k(1) = 0;  
x = a:h:b;  
y = zeros(1,M);  
y = x.^2.*(exp(x)-exp(1));  
error = zeros(1,M);  
  
%y' = (2./x) * y + x.^2.*exp(x);  
  
x_k = a;
```

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for k = 1:M
    y_k(k+1) = y_k(k) + h.* ((2./x_k) .* y_k(k) + x_k.^2 .* exp(x_k));
    x_k = x_k + h;
    error(k) = abs(y_k(k)-y(k));
end

y(M+1)
y_k(M+1)

fprintf('The biggest error of euler method is %f',max(error))
%max(error)%the error is a little big, unless you put the M very high.

%-----
%-----Modified Euler Method-----
% Use the second-order Talor expansion, import y''(x_k) and use the
% forward-difference formula to replace the y''(x_k).
clc
clear all
close all;

a = 1;
b = 2;
M = 20;
h = (b-a)/M;

p_k = zeros(1,M);
q_k = zeros(1,M);
y_k = zeros(1,M);
y_k(1) = 0;
x = a:h:b;
y = zeros(1,M);
y = x.^2.*(exp(x)-exp(1));
error_M = zeros(1,M);

x_k = a;
for k = 1:M
    p_k(k+1) = y_k(k) + h.* ((2./x_k) .* y_k(k) + x_k.^2 .* exp(x_k));

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    q_k(k+1) = y_k(k) + h.*(2./(x_k + h) .* p_k(k+1) + (x_k+h).^2 .*
exp(x_k + h));
    y_k(k+1) = 1/2.*(p_k(k+1)+q_k(k+1));
    error_M(k) = abs(y_k(k)-y(k));
    x_k = x_k + h;
end

y(M+1)
y_k(M+1)

fprintf('The biggest error of modified euler method is %f',max(error_M))

```

Euler Method->M	20	50	100	150	200
y	18.6831	18.6831	18.6831	18.6831	18.6831
y_k	18.0829	17.9656	18.3202	18.4403	18.5006
$Error(Max)$	0.577103	0.684459	0.354459	0.239083	0.180362

Modify Euler->M	20	50	100	150	200
y	18.6831	18.6831	18.6831	18.6831	18.6831
y_k	18.6782	18.6782	18.6818	18.6825	18.6828
$Error(Max)$	0.004725	0.004725	0.001228	0.000553	0.000313

It is easy to find that the modify Euler method is better than Euler method.